

Evaluating if a strategy pair is SPNE in the EcoPG stochastic game

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1 Definitions

To evaluate if a strategy pair is SPNE, we need to check if any agent has an incentive to make a one-step deviation from the strategy and then return to the same strategy, given the other agent is following the strategy. If there is no incentive to deviate for any agent, then the strategy pair is SPNE.

By incentive to deviate, we mean that the cumulative reward for the agent from deviating for a single step and then returning to the strategy is greater than the cumulative reward from following the strategy without deviation.

To estimate the cumulative rewards in either case (i.e., with deviation and without deviation), we write value functions for the agents.

In the EcoPG, the agents start in the prosperous state, and can transition to the degraded state with a probability dependent on the joint action. Once in the degraded state, they can either stay there or recover back to the prosperous state with some probability again dependent on their joint actions.

To derive the value functions for the prosperous state V_P and the degraded state V_D , we can use the Bellman equations.

Let the immediate payoff for the agent in the prosperous state be u , and the penalty associated with transitioning (or staying) in the degraded state be m . γ is the discount factor, p_c is the probability of collapse from the prosperous state to the degraded state, and p_r is the probability of recovery from the degraded state to the prosperous state.

Note that u , p_c , and p_r are all functions of the joint action of the agents, and hence should be written as $u(a_1, a_2)$, $p_c(a_1, a_2)$, and $p_r(a_1, a_2)$ respectively. However, for notational simplicity, we will write them without the arguments in the derivations below.

The value function in the prosperous state is

$$V_P = (1 - p_c) (V_P \gamma + u) + p_c (V_D \gamma + m) \quad (1)$$

Basically, $1 - p_c$ is the probability of staying in the prosperous state, in which case the immediate payoff is u and the discounted future payoff is γV_P . p_c is the probability of transitioning to the degraded state, in which case the immediate payoff is m and the discounted future payoff is γV_D .

Similarly, the value function in the degraded state is

$$V_D = m + \gamma (V_D (1 - p_r) + V_P p_r) \quad (2)$$

The immediate payoff will always be m in the degraded state, and the future payoff is a weighted average of the payoffs from staying in the degraded state (with probability $1 - p_r$) and recovering to the prosperous state (with probability p_r).

Rearranging the above equation, we get an expression for V_D

$$V_D = \frac{V_P \gamma p_r + m}{\gamma p_r - \gamma + 1} \quad (3)$$

Now, 3 can be substituted into 1 to get an expression for V_P in terms of the parameters of the model. Doing so and simplifying yields

$$\begin{aligned} V_P &= (1 - p_c)(V_P \gamma + u) + p_c \left(\frac{\gamma(V_P \gamma p_r + m)}{\gamma p_r - \gamma + 1} + m \right) \\ V_P &= \frac{\gamma m p_c p_r - \gamma p_c p_r u + \gamma p_c u + \gamma p_r u - \gamma u + m p_c - p_c u + u}{-\gamma^2 p_c - \gamma^2 p_r + \gamma^2 + \gamma p_c + \gamma p_r - 2\gamma + 1} \\ V_P &= \frac{m p_c (1 + \gamma p_r) + u (1 - p_c) (1 - \gamma (1 - p_r))}{(1 - \gamma) (1 + \gamma p_r - \gamma (1 - p_c))} \end{aligned} \quad (4)$$

We will use the last equation in all subsequent steps.

$$V_P = \frac{m p_c (1 + \gamma p_r) + u (1 - p_c) (1 - \gamma (1 - p_r))}{(1 - \gamma) (1 + \gamma p_r - \gamma (1 - p_c))} \quad (5)$$

2 Comparison of Cumulative Rewards with and without Deviation

Now, we compare the cumulative rewards for the agent from deviating for a single step and then returning to the strategy, with the cumulative reward from following the strategy without deviation.

Without deviation, the cumulative reward is simply V_P as derived above.

With deviation, we need to consider the consequences of a single step deviation. Let us say the agent deviates at time step $t = 0$. The immediate payoff from deviation is u^0 , and the probability of collapse due to deviation is p^0 .

So, the cumulative reward from deviation at $t = 0$ can be written as:

$$V_P^{(0)} = (1 - p^0)(u^0 + \gamma V_P^{(1)}) + p^0(m + \gamma V_D^{(1)}) \quad (6)$$

In 6, $V_P^{(1)}$ is the cumulative reward from time step $t = 1$ onwards, given that the agent deviated at $t = 0$ and then returned to the strategy. Likewise, $V_D^{(1)}$ is the cumulative reward from time step $t = 1$ onwards in the degraded state, given that the agent deviated at $t = 0$ and then returned to the strategy. The above equation is the general form of the SPNE equation. Thus, if a strategy is SPNE, then the following condition must hold:

$$V_P^{(0)} \leq V_P \quad (7)$$

$$\boxed{(1 - p^{(0)})(u^0 + \gamma V_P^{(1)}) + p^{(0)}(m + \gamma V_D^{(1)}) \leq \frac{m p_c (1 + \gamma p_r) + u (1 - p_c) (1 - \gamma (1 - p_r))}{(1 - \gamma) (1 + \gamma p_r - \gamma (1 - p_c))}} \quad (8)$$

2.1 SPNE inequality in different info conditions

We look at how how $V_P^{(0)}$ differs across the different information conditions.

2.1.1 Only Ecological Information:

Under the only ecological information condition, the agents can distinguish between the prosperous and dedgraded states, but they play memory-0 strategies. So, a one step deviation will affect only the payoff and the probability of collapse at $t = 0$, but this effect does not carry on the future steps.

So, we can write the cumulative reward from deviation as:

$$V_P^{(0)} = (1 - p^{(0)})(u^0 + \gamma V_P) + p^{(0)}(m + \gamma V_D) \quad (9)$$

This is simply 6 with $V_P^1 = V_P$ and $V_D^1 = V_D$.

2.1.2 Complete Information

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If agents play memory-1 strategies in addition to having to knowledge of the ecological state, then a one step deviation can have consequences beyond just the time-step in which the deviation occurs.

For example, consider a strategy like WSLs. If both agents are playing WSLs and cooperating from the beginning, then if one agent deviates at $t = 0$ by defecting, then at $t = 1$, both agents will defect. However, at $t = 2$, both agents will again cooperate. So, the effect of deviation lasts for one more time step after the deviation step. So, we need to compute V_P^1 and V_D^1 separately, since they will differ from V_P and V_D respectively. But V_P^2 and V_D^2 will be the same as V_P and V_D respectively, since after $t = 2$, the effect of deviation is gone and both agents are back to cooperating.

$$V_P^{(0)} = (1 - p_c^{(0)}) (u + \gamma V_P^{(1)}) + p_c^{(0)} (m + \gamma V_D^{(1)}) \quad (10)$$

$$\begin{aligned} V_P^{(1)} &= (1 - p_c^{(1)}) (u + \gamma V_P^{(1)}) + p_c^{(1)} (m + \gamma V_D^{(1)}) \\ V_D^{(1)} &= m + \gamma ((1 - p_r^{(1)}) V_D^{(1)} + p_r^{(1)} V_P^{(1)}) \end{aligned} \quad (11)$$

If needed in GT like strategies, the effect of deviation can be permanent, and the agents that initially cooperated keep defecting forever after the deviation. In this case, we need to compute $V_P^{(2)}$ and $V_D^{(2)}$ separately as well. However, after $t = 2$, the effect of deviation is permanent, so $V_P^{(3)} = V_P^{(2)}$ and $V_D^{(3)} = V_D^{(2)}$.

$$V_P^{(0)} = (1 - p_c^{(0)}) (u + \gamma V_P^{(1)}) + p_c^{(0)} (m + \gamma V_D^{(1)}) \quad (12)$$

$$\begin{aligned} V_P^{(1)} &= (1 - p_c^{(1)}) (u + \gamma V_P^{(1)}) + p_c^{(1)} (m + \gamma V_D^{(1)}) \\ V_D^{(1)} &= m + \gamma ((1 - p_r^{(1)}) V_D^{(1)} + p_r^{(1)} V_P^{(1)}) \end{aligned} \quad (13)$$

$$\begin{aligned} V_P^{(2)} &= (1 - p_c^{(2)}) (u + \gamma V_P^{(2)}) + p_c^{(2)} (m + \gamma V_D^{(2)}) \\ V_D^{(2)} &= m + \gamma ((1 - p_r^{(2)}) V_D^{(2)} + p_r^{(2)} V_P^{(2)}) \end{aligned} \quad (14)$$

2.1.3 No Information

Under the no information condition, the agents cannot distinguish between the prosperous and degraded states. So, from the POV of the agents, there is no V_P and V_D , but a single value function $V_{\text{no info}}$ which is the same for both states.

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2.1.4 Only Social Information

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